

Shirley Katherine Maldonado

Data Analyst | Data Scientist

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PROFILE

Data analyst and data scientist with a background in Astronomy, hands-on experience in the retail sector, and C1-level English proficiency. I work primarily in Python, R, SQL, and Excel for data extraction, transformation, and modeling, and in Power BI to communicate results to business audiences. My approach combines statistical rigor with the ability to translate technical findings into actionable decisions. [Download link](#)

TECHNICAL SKILLS

Programming:	Python, R, SQL.
Databases:	MySQL, SAS.
Visualization:	Power BI, Matplotlib, Seaborn, Qualtrics.
Libraries:	Pandas, NumPy, scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn.
Machine Learning:	Histogram Gradient Boosting, Regression, Classification, Decision Trees, SHAP values, permutation importance for interpretability.
Statistical Methods:	PCA, Multiple Correspondence Analysis (MCA), causal discovery.

WORK EXPERIENCE

Market Research & Data Science Intern	Grupo Éxito
– Developed an omnichannel NPS predictive model using Gradient Boosting (R^2 between 0.63 and 0.79 on the test set), identifying on-time product delivery as the primary driver of customer satisfaction and loyalty. Results were presented to the team and adopted as an operational prioritization framework. – Optimized the customer experience survey through PCA and correlation analysis, identifying redundant or low-variance items (such as available payment methods and staff knowledge) whose removal reduces response burden without loss of explanatory power (variance explained by Moment of Truth: 75–85%). – Designed a brand reputation analysis using ordinal logistic regression (instead of the standard permutation importance approach) to quantify the impact of each variable on consumer trust. This methodological choice, proposed by me and validated by the team, yielded directly interpretable weights that were aggregated at the driver level and translated into a performance-vs.-impact chart consistent across audiences and levels of analysis. – Analyzed Nielsen data for competitive benchmarking and supported the Data Science team in brand perception metrics.	

PROJECTS

Time Series Analysis and Electricity Price Forecasting
Academic team project in which we processed historical data from XM and built LSTM models in TensorFlow/Keras to forecast the kWh price in the short and medium term.
Database Design for Disaster Risk Management in Antioquia
Capstone project of the Betek Data Analytics Bootcamp, presented at the company’s Open House to an external audience. I designed the relational model in MySQL, conducted exploratory data analysis in Python, and built a Power BI dashboard covering the historical record of disasters and risk zones in the department of Antioquia.

EDUCATION

Mar 2019 – Mar 2026	B.Sc. in Astronomy	Universidad de Antioquia
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CONTINUING EDUCATION

Oct 2024 – Dec 2024	Power BI and Data Visualization Course	Universidad de Antioquia
Nov 2023 – May 2024	Data Analytics Bootcamp	Betek
Nov 2023 – Feb 2024	Diploma in Data Analysis and Machine Learning with Python	Universidad de Antioquia
Nov 2022 – Dec 2022	Fundamentals of Data Analytics	DS4A